NPLify · P0 Technical Baseline

Ledger Posting Design & Chart of Accounts · Draft v5.9

Numbered with the ERD edition it is built against

New XP Technologies Limited

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Status: Built against ERD & Data Model Draft v5.9 (decisions D1–D46), which carries every decision of this document as D46. Every worked example was produced by a small posting engine that refuses any transaction whose debits and credits do not match per currency and per owner, so the figures add up by construction. Supersedes Draft v1.1 (ERD v5.2); the document is numbered with the ERD edition it is built against.

Audience: NPL Finance and Management, and the New XP engineering team. Written in plain language; the account codes and field names are the ones in the ERD, with NPL's vocabulary (settlement, partner entity, markup share, sender pays / receiver pays, fee on market / fee in rate, agent rate).

> **What changed since v1.1.** The chart of accounts grows with the v5.3–v5.6 purposes (EARN_REBATE, SHARE_RECEIVABLE, EXP_ROUNDING, EXP_LOSS) and with the purposes this design put into the ERD as D46 (Section 24): four variance accounts in place of one, loss receivables and the unfunded make-good pair, the reseller revenue and cost accounts, and FX_RESULT for NPL's own currency exchanges. Rebates and markup shares are accrued at conversion. A loss is booked to NPL in full and recovered share by share. NPL's earnings may be kept in the unconverted balance and offset between partners (Pattern N). The NPL-GR project keeps a full reseller P&L inside NPLify: the vendor and resale invoices are posted in the invoice currency, the remittance deal posts like any client project with NPL-GR as the client, and the forex result of instalments lands on the remittance margin. New worked patterns: same-currency pass-through, round-up rounding, bridge-wallet hop, party-retains share with rebate, cutoff and honoured-rate variances, loss event with recoveries, the reseller deal with its two invoices and instalments. v5.9 corrects the wording left behind by D46 and shows T41 as one recovery posting per share.

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1 · What the ledger is for, in one paragraph

NPL coordinates other people's money and, in one project, is also its own client: NPL-GR resells goods and pays for the remittance like any customer. The ledger answers, at any moment and for any past moment, three questions: **where is the money** (at which partner, sent and awaiting confirmation, in one of NPL's wallets), **whose money is it** (the client's, a sender's, or NPL's), and **what did each movement cost or earn** (NPL's margin and shares of it, rebates, partner cost, bank and network fees, rounding, losses, and the four kinds of variance). It does this with ordinary double-entry bookkeeping: every movement is a set of debits and credits that balance, nothing is ever edited or deleted, and every balance anyone sees is a sum over those entries.

2 · Rules that every posting follows

1. **One transaction per event.** A collection, a conversion, a balance conversion, a settlement, a settlement return, a confirmation, a bank-fee event, a reroute, a markup-share or rebate accrual or settlement, a loss event or recovery, and a manual adjustment each produce exactly one `LEDGER_TRANSACTION` that points at its source record (`source_type`, `source_id`). A source row with a lifecycle — a share or rebate accrual, a loss recovery — produces one transaction per state change, each naming the same row; a full confirmation and a hop release are events that post nothing (rule 12).
2. **Balanced per currency and per owner.** Within one transaction the debits equal the credits in each currency separately, and separately again for client-tagged and company-tagged lines. A conversion therefore has an in-currency part and an out-currency part that each balance on their own; no exchange-rate plug is ever needed, and nothing silently moves between the client's books and NPL's.
3. **Ownership is a tag on every line** (`ownership_tag` client / company) and the account code says the same (`CL.` / `CO.`). Ownership moves at named events only (invariant 16). Client → company: margin recognition at conversion; the client side of an earnings offset; a forex true-up gain on an invoice line; the client's share of a loss. Company → client: a fee NPL absorbs or covers, a round-up, a shortfall top-up, a bounce make-whole, a loss make-good; the company side of an earnings offset; a forex true-up loss. Each transaction balances per owner, so these movements are always explicit lines.
4. **A cost component on every line** (`cost_component`): principal, earnings, rebate, partner_cost, bank_fee, network_fee, variance, share, rounding, fx_timing, cutoff_timing, loss, and **resale** for the reseller project. Reports of margin, cost and variance are sums over this tag; no figure is stored.
5. **Accruals post with the conversion.** NPL's margin, each markup-share party's part and the partner rebate expected on the leg are all recognised at conversion, as separate transactions that name the same deal; the monthly statement settles them and any difference is a true-up, never a rewrite.
6. **Variances are booked by kind**, to four accounts: VAR_CONVERSION (the partner did something other than expected), VAR_CUTOFF (D29), VAR_RATE_HONOUR (D45), VAR_FX_TIMING (D24 and the own desk). Each can be a gain or a loss; each line names the leg or line it arose on.
7. **A loss is NPL's first.** A LOSS_EVENT posts the whole amount to EXP_LOSS and makes the client whole in the same transaction; each share the partner, the sender or the client agrees to bear is a separate, later recovery posting.
8. **NPL-GR is the client of its own remittance project.** Its reseller books are two postings per deal in the invoice currency, one per invoice, and a claim on the remittance side; the remittance deal posts like any client project; the forex result of paying the vendor in instalments is a true-up to VAR_FX_TIMING against NPL's pool, so the reseller revenue is never diluted.
9. **Earnings may be kept in the collection currency and offset between partners.** A conversion may keep the deal's margin unconverted (residual_in) and an approved EARNINGS_OFFSET may swap NPL's margin at one partner for the same value of unconverted client money at another, at a stated rate; the client's entitlement never changes, and NPL's exchange of one currency for another is recorded in FX_RESULT (Pattern N).
10. **The views are sums over postings.** PROJECT_BALANCE is the balance of the DUE account per partner; GROUP_ENTITLEMENT is the sum of PAYABLE postings by receiver_group_id, which every promise line carries; neither is computed from deals (D46 amendment).
11. **Append-only.** A correction is a reversing transaction that names what it reverses (`reversal_of_id`) followed by the correct posting (invariant 2).
12. **Amounts stay in their own currency.** A converted total (volumes, earnings, balances, custody in the reporting currency) is a REPORTING_VALUE: computed at read time from the latest platform-pair market rate, carrying that rate and its time, flagged when stale, and never posted, quoted or settled (invariant 38).

13. **Sent is discharged.** A settlement line discharges the promise when the partner releases the money; the receiver's confirmation posts nothing, a short delivery re-raises the promise for the missing part, and a return re-raises it for what came back. What is sent and unconfirmed is the line's outcome (SETTLEMENT_LINE.outcome sent), not a jar; every released line ends confirmed, short or returned.
14. **Rounding has an owner.** Bank transfers go out exact, cents included (amount_rounding exact); rounding to a unit applies where the pair says so, in practice to cash hand-overs. A dropped fraction stays client money in the balance; an amount added by rounding up is NPL's cost in EXP_ROUNDING (invariant 24).

3 · Chart of accounts

Account codes read `OWNER.HOLDER.CURRENCY.PURPOSE`. The owner is `CL` for client funds or `CO` for company. The holder says where the money is or whom it concerns: a partner code such as `ALI`, `JETON` or `AQN` (a `PARTNER_CONFIG`, so always within one project), `OWNDESK` for the own-desk configuration of the reseller project, `OWN` for NPL's own wallets, a receiver, a sender, a markup-share party, `PROJECT` for the project as a whole, or `NONE` for income, expense and clearing accounts. One account exists per combination actually used; accounts are created on first posting (`LEDGER_ACCOUNT`: code, ownership, holder_type, holder_id, currency, purpose, kind). Purposes marked **v5.7 (D46)** entered the ERD with this edition; Section 24 lists the decision.

Code	Kind	Since	What the balance means
`CL.<PARTNER>.<CCY>.COLLECTED`	asset	v5.2	Client money received at this partner and not yet converted (in the collection currency). In the reseller project NPL-GR is the client, so its collections are CL as well.
`CL.OWN.<CCY>.HELD`	asset	v5.2	Client money in one of NPL's own wallets: an approved reroute, or a collection bridge waiting to be forwarded (D40). Zero otherwise (invariant 3).
`CL.<PARTNER>.<CCY>.DUE`	asset	v5.2	Converted client money resting at this partner, waiting to be settled: one account per project × partner × currency, the PROJECT_BALANCE.
`CL.<PARTNER> / OWN.<CCY>.INTRANSIT`	asset	**retired v5.7 (D46)**	Retired in v5.7 (NPL, 11 October): once money has left the partner it is out of NPL's hands, so the promise is discharged on release and re-raised only by a return or a short delivery. What is sent and not yet confirmed is read from SETTLEMENT_LINE.outcome, not from a jar.
`CL.<RECEIVER>.<CCY>.SHORTFALL`	asset	v5.2	Client money delivered short to a receiver and still to be made good at the next settlement.
`CL.PROJECT.<CCY>.PAYABLE`	liability	v5.2	What NPL must deliver to the project's receivers, in total per currency. The split by receiver group is the GROUP_ENTITLEMENT view, not an account.
`CL.<SENDER>.<CCY>.CREDIT`	liability	v5.2	Owed back to a sender: an overpayment waiting to be refunded or applied to the next deal (SENDER_CREDIT).
`CL.PROJECT.<CCY>.MAKEGOOD`	asset	**v5.7 (D46)**	Client money NPL has undertaken to replace after a loss but has not yet placed at a partner. Appears only when NPL's pool at the partner cannot fund the make-good at once; cleared when NPL's funds land (Pattern L).
`CO.<PARTNER>.<CCY>.POOL`	asset	v5.2	NPL's own money resting at this partner: margin captured at conversion (EARNINGS_RECEIVABLE), plus anything NPL transferred there. Settled on the partner's fee cycle or netted on a reroute.

Code	Kind	Since	What the balance means
`CO.OWN.<CCY>.WALLET`	asset	v5.2	NPL's own money in its own wallets: recovered dues, rebates and shares received, network fees paid.
`CO.<PARTNER>.<CCY>.REBATE_RECEIVABLE`	asset	v5.2	Rebate a partner owes NPL on NPLify deals, accrued at conversion and reconciled against the partner's monthly statement (D27).
`CO.<PARTY>.<CCY>.SHARE_RECEIVABLE`	asset	v5.3	NPL's remainder of the markup owed by a party that holds the converted funds (settlement_mode party_retains, D26); reconciled monthly.
`CO.<PARTY>.<CCY>.SHARE_PAYABLE`	liability	v5.2	Markup share owed by NPL to a markup-share party (settlement_mode npl_pays, D26).
`CO.<PARTNER> / <SENDER>.<CCY>.LOSS_RECEIVABLE`	asset	**v5.7 (D46)**	The partner's or the sender's agreed share of a loss event, owed to NPL after NPL booked the whole loss first (Pattern L).
`CO.NONE.<CCY>.LOSS_PAYABLE`	liability	**v5.7 (D46)**	The unfunded part of a make-good: what NPL still has to put in after a loss. Mirrors CL.PROJECT.<CCY>.MAKEGOOD and is cleared by the same transfer.
`CO.<VENDOR>.<CCY>.VENDOR_PAYABLE`	liability	**v5.7 (D46)**	NPL-GR's reseller books: what NPL-GR owes a vendor on a vendor invoice, in the invoice currency; cleared as the remittance side's payout lines discharge the invoice (Pattern M).
`CO.<CUSTOMER>.<CCY>.RESALE_RECEIVABLE`	asset	**v5.7 (D46)**	NPL-GR's reseller books: what a reseller customer owes on a resale invoice, in the invoice currency; cleared when the customer pays the remittance side (Pattern M).
`CO.PROJECT.<CCY>.REMIT_CLAIM`	asset	**v5.7 (D46)**	NPL-GR's reseller books: what the remittance side holds for NPL-GR, in the invoice currency — the mirror, in EUR, of the remittance project's PAYABLE to its client NPL-GR (Pattern M).
`CO.NONE.<CCY>.EARN_GROSS`	income	v5.2	Margin recognised at conversion (sender and receiver fee parts), before partner cost. Markup shares are debited here, so the balance is NPL's margin after shares.
`CO.NONE.<CCY>.EARN_REBATE`	income	v5.3	Rebate income, kept apart from EARN_GROSS and never in a markup-share base (D27).
`CO.NONE.<CCY>.RESELL_REVENUE`	income	**v5.7 (D46)**	NPL-GR's reseller books: the resale invoice to the customer at retail price, in the invoice currency, recognised when issued (Pattern M).
`CO.NONE.<CCY>.RESELL_COST`	expense	**v5.7 (D46)**	NPL-GR's reseller books: the vendor invoice at NPL's preferential price, in the invoice currency, recognised when received. Revenue – cost is the game reseller revenue, fixed on issue and never touched afterwards (Pattern M).
`CO.NONE.<CCY>.EXP_PARTNER`	expense	v5.2	Partner cost where the partner states it separately (fee_on_market). Zero by construction for fee_in_rate partners and the own desk.
`CO.NONE.<CCY>.EXP_BANKFEE`	expense	v5.2	Bank and payout fees NPL absorbs: gross-ups, shortfall top-ups, bounce fees, covered fees (D41).
`CO.NONE.<CCY>.EXP_NETWORK`	expense	v5.2	Crypto network fees NPL absorbs, in USDT: a refund, a bridge hop, a forwarded reroute. Zero when none applies.
`CO.NONE.<CCY>.EXP_ROUNDING`	expense	v5.3	The amount added when a payout is rounded up to the pair's unit (D25).

Code	Kind	Since	What the balance means
`CO.NONE.<CCY>.EXP_LOSS`	expense	v5.3	Loss events. The whole loss is booked here first; each recovered share is credited back as it is agreed (Pattern L).
`CO.NONE.<CCY>.VAR_CONVERSION`	expense	**v5.7 (D46)**	Partner converted differently from the engine's expectation (partner_calculates), and basis variance on a reroute. Gain or loss. Was VARIANCE.
`CO.NONE.<CCY>.VAR_CUTOFF`	expense	**v5.7 (D46)**	Rate stamped at lock versus the rate actually used when the conversion ran after the partner's cutoff; NPL absorbs it (D29).
`CO.NONE.<CCY>.VAR_RATE_HONOUR`	expense	**v5.7 (D46)**	An expired quote's rate honoured by an approved override versus the partner's rate on the day of conversion; NPL's cost, chosen (D45).
`CO.NONE.<CCY>.VAR_FX_TIMING`	expense	**v5.7 (D46)**	The forex result of discharging an obligation stated in one currency with money held in another: each vendor instalment and the margin line of a resale invoice (D24). Gain or loss, always on the remittance margin, never on the reseller revenue.
`CO.NONE.<CCY>.FX_RESULT`	expense	**v5.7 (D46)**	NPL's own currency exchanges, a profit-and-loss account per currency: currency NPL sold is a debit, currency it bought a credit — an earnings offset, a pool squared from USDT, a company balance settled in another currency. Its reporting-currency value is NPL's realised currency result; it needs no later closing (D46 amendment: replaces FX_CLEARING, an asset that could never be closed under per-currency balancing).

Debit-normal accounts (assets, expenses) grow with debits; credit-normal accounts (liabilities, income) grow with credits. A variance account with a credit balance is a gain. `ADVANCE` does not exist because NPL never advances money to a receiver (invariant 16). The former single `VARIANCE` purpose is replaced by the four VAR_ accounts; the former referral payable is `SHARE_PAYABLE` (D37).

4 · Cost components, owners and sources

Cost component	Used on	Meaning
principal	client money moving	The amount itself: collected, converted, released, confirmed, returned.
earnings	POOL, EARN_GROSS, WALLET on dues recovery; the client-owned DUE, PAYABLE and COLLECTED lines of an earnings offset or a retained margin (T54, T55)	NPL's margin (sender and receiver fee parts) and where it rests.
partner_cost	EXP_PARTNER	A fee_on_market partner's stated fee, deducted at transaction level.
rebate	REBATE_RECEIVABLE, EARN_REBATE, WALLET	A partner's rebate on NPLify deals: accrual, receipt, true-up.
share	EARN_GROSS (debit), SHARE_PAYABLE, SHARE_RECEIVABLE	A markup-share party's part of the net markup and its settlement.
bank_fee	EXP_BANKFEE, SHORTFALL, POOL, DUE	Bank and payout fees: waived, carried forward, absorbed, covered, bounce.
network_fee	EXP_NETWORK, WALLET, COLLECTED	Crypto network fees on refunds, hops and forwards, stated in USDT; zero, and no line, when none applies.

Cost component	Used on	Meaning
rounding	DUE, EXP_ROUNDING, POOL	The signed residual of the pair's rounding option.
variance	VAR_CONVERSION, VAR_RATE_HONOUR	Expected versus actual, by kind.
cutoff_timing	VAR_CUTOFF	Rate at lock versus rate used after the partner's cutoff.
fx_timing	VAR_FX_TIMING	Obligation priced on one day, discharged on another.
loss	EXP_LOSS, LOSS_RECEIVABLE, POOL, WALLET, COLLECTED, PAYABLE	A loss event, its make-good and its recoveries.
resale	RESELL_REVENUE, RESELL_COST, VENDOR_PAYABLE, RESALE_RECEIVABLE, REMIT_CLAIM, FX_RESULT	NPL-GR's reseller books: the two invoices and the claim on the remittance side.

Transaction sources (`LEDGER_TRANSACTION.source_type`): Collection, Conversion, BalanceConversion, Settlement, SettlementReturn, Confirmation, BankFeeEvent, Reroute, MarkupShare, Rebate, LossEvent, **LossRecovery** for each recovered share, Adjustment, **Invoice** for the reseller books and **EarningsOffset** for the cross-partner offset. Section 22 lists what each one posts; Section 24 names the source record (LOSS_RECOVERY) a loss recovery points at.

5 · Reading a posting table

Each worked example shows one transaction as a table. *Debit* and *Credit* are the amounts in bookkeeping's two columns; *In / Out* says the same thing in one word: every transaction moves value from somewhere to somewhere, a **debit** marks the account the value goes **to** (IN) and a **credit** the account it comes **from** (OUT). The word after the dot says what kind of account it is. **money** is a place where money sits (a partner, a wallet, a balance): IN means money arrives there, OUT that it leaves. **promise** is what NPL has undertaken to deliver: OUT means NPL now holds money on the strength of a promise (the promise is the source), IN means the promise is being fulfilled. **earned** is NPL's margin or income: OUT means earnings are the source of money that lands somewhere, usually NPL's pool at a partner. **cost** is what NPL bears: IN means value goes to a cost. A money or cost account is full when more has gone in than out; a promise or earned account is full when more has come out of it than gone in, because those two are sources by nature; the balance tables show the right figure either way. A reader who ignores the Debit and Credit columns and reads only Account, In / Out and Why will follow every example. *Owner* is the ownership tag; *Cost component* is the breakdown tag; *Why* says in plain words what that line records. The running example is project Evo: Sender A, Receiver X (receiver group A, the default group) with entities Entity X1 and Entity X2; partner Ali (fee_in_rate: its cost is inside its rate), partner Jeton (fee_on_market + 1.25 %, Evo's Jeton fee) and partner Aquanow (fee_on_market + 0.80 %, rebate 0.3 % of the amount). Rates: market 0.9000 EUR per USDT, Ali 0.8910 (Ali's cost is assumed at 1 % inside its rate; the real figure changes daily). Fee structure: rate_basis market (the sender is priced on the market rate and Ali converts at its own), fixed, 2.00 % in total: this sender type pays 1.40 % (Evo's other type pays 1.00 %) and the receiver always pays 0.60 %. The arithmetic is in the Calculation Specification; the headline figures for deal 1 are:

Figure	Deal 1 (10,000 USDT at Ali)	How
Sender rate	0.88740	market 0.9000 × (1 – 1.40 %): the sender's part is taken off the rate
Gross out	8,874.00 EUR	10,000 × sender rate

Figure	Deal 1 (10,000 USDT at Ali)	How
Receiver pays	54.00 EUR	10,000 × market × 0.60 %: both parts are on the principal, so client net = 10,000 × market × (1 – 2.00 %)
Client net (entitlement)	8,820.00 EUR	gross out – receiver part
Converted at Ali	8,910.00 EUR	10,000 × 0.891 (Ali's own rate; its spread against market is 90.00 EUR)
NPL earnings	90.00 EUR	converted – client net = fee parts 180.00 (sender 126.00 + receiver 54.00) – Ali's spread 90.00
Paid out	8,820.00 EUR	amount_rounding exact: a EUR bank transfer carries the cents; only cash hand-overs round to a unit

6 · Pattern A — direct collection, conversion and accumulation

The everyday case: the sender pays, the partner converts, the client balance at that partner grows until it is settled. Collect-first and quote-first deals post identically; only the moment pricing is fixed differs. A quote extended any number of times (D45) still posts once, at conversion.

T1 · Collection of deal 1: 10,000 USDT arrive at Ali's USDT wallet endpoint (COLLECTION_RECEIVING_ENDPOINT, kind wallet) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	client money now sits at Ali, unconverted; the account is per partner, the endpoint only says which vehicle's wallet
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	NPL owes this onward on the client's behalf

T2 · Conversion of deal 1 at Ali (fee_in_rate): 10,000 USDT → 8,910.00 EUR at 0.891 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	the USDT obligation is discharged by the conversion
CL.ALI.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	the USDT left the collected position
CL.ALI.EUR.DUE	8,820.00		**IN** · money	EUR	client	principal	client net, waiting to be settled (EUR pairs round exact: cents are paid out)
CO.ALI.EUR.POOL	90.00		**IN** · money	EUR	company	earnings	NPL's margin, physically still at Ali (EARNINGS_RECEIVABLE)
CL.PROJECT.EUR.PAYABLE		8,820.00	**OUT** · promise	EUR	client	principal	owed to the receiver group, now in EUR
CO.NONE.EUR.EARN_GROSS		90.00	**OUT** · earned	EUR	company	earnings	margin recognised: the client's claim shrinks from 10,000 USDT to its EUR entitlement, the difference is NPL's

Ownership changes here and only here: 90.00 EUR becomes company money, because the client's claim is re-denominated at the sender's price while the partner delivers at its rate. The sender is priced on the market rate (rate_basis market: the fee is disclosed) and Ali converts at its own rate, so Ali's cost — 90.00 EUR against market, inside its rate (fee_in_rate) — comes out of NPL's 2.00 % and no partner-cost line exists; NPL keeps the 90.00 EUR. Under rate_basis agent_rate (a calculated rate, breakdown not disclosed) the sender's rate would be Ali's rate less the fee instead, and NPL would keep the whole 2.00 %; the postings have the same shape. Evo's other sender type pays 1.00 % instead of 1.40 %: a second fee structure (or a FEE_OVERRIDE with its own split, D20), same postings, smaller sender part. Nothing differs for a leg split across two fee tiers (D21): the fee parts are the blended sum and the posting is identical. The customer-facing rate and fee (D22) are print-only and never post.

Balances after deal 1

Account	Ccy	Balance	Meaning
CL.ALI.USDT.COLLECTED	USDT	0.00	nothing left unconverted
CL.ALI.EUR.DUE	EUR	8,820.00	client balance at Ali (PROJECT_BALANCE)
CL.PROJECT.EUR.PAYABLE	EUR	8,820.00	owed to receivers; equals group entitlement

Account	Ccy	Balance	Meaning
CO.ALI.EUR.POOL	EUR	90.00	NPL's margin resting at Ali
CO.NONE.EUR.EARN_GROSS	EUR	90.00	margin recognised to date

T3a · Same-currency collection at Ali: 5,000 EUR by bank transfer (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.EUR.COLLECTED	5,000.00		**IN** · money	EUR	client	principal	
CL.PROJECT.EUR.PAYABLE		5,000.00	**OUT** · promise	EUR	client	principal	

T3 · Same-currency pass-through (D33): the 5,000 EUR converted at rate 1 — Evo's EUR → EUR fee 2.80 % (sender pays 1.40 %, receiver pays 1.40 %) captured, client net 4,860.00 EUR (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.EUR.PAYABLE	5,000.00		**IN** · promise	EUR	client	principal	the obligation is re-stated at the sender's price
CL.ALI.EUR.COLLECTED		5,000.00	**OUT** · money	EUR	client	principal	
CL.ALI.EUR.DUE	4,860.00		**IN** · money	EUR	client	principal	client net, available to settle
CO.ALI.EUR.POOL	140.00		**IN** · money	EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,860.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		140.00	**OUT** · earned	EUR	company	earnings	the fee is earned at conversion even though no currency changes

A pair with the same currency on both sides converts at rate 1 like any other pair (invariant 32): the in-leg and the out-leg happen to be in one currency, and the balance arithmetic is unchanged.

7 · Pattern B — part collection

A deal may be collected in parts, by any method; the obligation grows with each part and the conversion waits for the last one (or for the partner's cash count).

T4 · Collection of deal 2, part 1 of 2: 4,000 USDT source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	4,000.00		**IN** · money	USDT	client	principal	first part
CL.PROJECT.USDT.PAYABLE		4,000.00	**OUT** · promise	USDT	client	principal	

T5 · Collection of deal 2, part 2 of 2: 1,000 USDT (next day) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	1,000.00		**IN** · money	USDT	client	principal	second part completes the deal
CL.PROJECT.USDT.PAYABLE		1,000.00	**OUT** · promise	USDT	client	principal	

T6 · Conversion of deal 2 at Ali: 5,000 USDT → 4,455.00 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	5,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		5,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	4,410.00		**IN** · money	EUR	client	principal	
CO.ALI.EUR.POOL	45.00		**IN** · money	EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,410.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		45.00	**OUT** · earned	EUR	company	earnings	

Parts are posted as they arrive; the conversion is one transaction once the deal is fully collected. A cash part is verified only after the partner's count (counted_at, D30) and posts then; the cash amount itself was rounded to the pair's cash unit at quote (cash_rounding, D25), so no residual arises on the collection side.

8 · Pattern C — settlement, confirmation, bank fees, shortfalls, rounding

A settlement drains a partner's holding into one or many lines; each line is confirmed by its receiver in full or short. The bank-fee treatment of the project or of a FEE_DECISION decides who bears a fee the bank took.

T7 · Settlement from Ali: 18,090.00 EUR released in two lines (Entity X1 8,000.00, Entity X2 10,090.00); the promise is discharged on release source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.EUR.PAYABLE	18,090.00		**IN** · promise	EUR	client	principal	once the money has left Ali it is no longer in NPL's hands, so NPL no longer owes it; a return or a short delivery re-raises the promise
CL.ALI.EUR.DUE		18,090.00	**OUT** · money	EUR	client	principal	both lines leave the client's balance at Ali; EUR bank transfers carry the cents (amount_rounding exact)

The client balance at Ali drops to zero (invariant 4, settle to the residual: a pair rounded to a unit would leave only the rounding residual). Both lines pay entities of the same receiver and each names the leg it settles in deal_id, so no offset arises. The lines now have outcome sent; what is sent and not yet confirmed is read from SETTLEMENT_LINE.outcome, not from a jar. The transaction posts when the partner releases the money, which for a bank rail is no earlier than execution_due_date (approval before the partner's cutoff → same day, after it → next business day; weekends and HOLIDAY rows skipped, D29, D34).

T8 · Confirmation, line 1 full: Entity X1 received 8,000.00 EUR source: Confirmation

No posting. No posting: the promise was discharged when the money left Ali (T7). The line's outcome becomes confirmed, and the control that every released line ends as confirmed, short or returned is read from the lines.

T9 · Confirmation, line 2 short: Entity X2 received 10,040.00 EUR, the bank took 50.00 (treatment carry_forward) source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.RECEIVER_X.EUR.SHORTFALL	50.00		**IN** · money	EUR	client	bank_fee	still owed to the receiver, topped up at the next settlement
CL.PROJECT.EUR.PAYABLE		50.00	**OUT** · promise	EUR	client	bank_fee	the promise is re-raised for the part that did not arrive

With treatment *waived* nothing posts: the client bears the fee and its claim stays discharged. With *absorbed* the fee is expected up front and funded as in T10. A fee *covered* by a FEE_DECISION (D41: Finance recommends, Management approves) posts exactly as absorbed.

T10 · Shortfall top-up funded from NPL's pool at Ali (BANK_FEE_EVENT applied to the next settlement) source: BankFeeEvent

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.EXP_BANKFEE	50.00		**IN** · cost	EUR	company	bank_fee	NPL bears the cost
CO.ALI.EUR.POOL		50.00	**OUT** · money	EUR	company	bank_fee	paid out of NPL's money resting at Ali
CL.ALI.EUR.DUE	50.00		**IN** · money	EUR	client	bank_fee	the 50 EUR is client money again, in balance at Ali
CL.RECEIVER_X.EUR.SHORTFALL		50.00	**OUT** · money	EUR	client	bank_fee	the shortfall is cleared

The 50 EUR now sits in DUE and goes out with the next settlement like any other balance. Company → client transfers like this one are the only way company money becomes client money (besides a loss make-good, Pattern L). **If NPL's EUR pool at Ali cannot cover the 50** (its earnings there may be in another currency, or already taken out), the posting is the same and the pool goes negative: Ali has fronted the 50 out of the one balance it holds. NPL squares it at the next conversion by having its own USDT converted along with the client's, posted as in T24a.

Balances after settlement and confirmation

Account	Ccy	Balance	Meaning
CL.ALI.EUR.DUE	EUR	50.00	the topped-up 50 EUR, out with the next settlement
CL.PROJECT.EUR.PAYABLE	EUR	50.00	the 50 EUR still owed to Receiver X
CL.RECEIVER_X.EUR.SHORTFALL	EUR	0.00	cleared
CO.ALI.EUR.POOL	EUR	225.00	NPL's margin at Ali after funding the top-up
CO.NONE.EUR.EXP_BANKFEE	EUR	50.00	bank fees NPL bore

T11 · Conversion for a cash payout with round-up (D25): 3,000 USDT → 96,810.00 THB at 32.27 (market 32.6); client net 95,844.00, handed over in cash as 96,000.00 (amount_rounding round_up_n, rounding_unit 1,000 — the payout option; cash_rounding is the collection side)

source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	(collection of the 3,000 USDT assumed posted as T1)
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.THB.DUE	95,844.00		**IN** · money	THB	client	principal	client net as priced
CL.ALI.THB.DUE	156.00		**IN** · money	THB	client	rounding	the amount added by rounding up: client money from now on
CO.ALI.THB.POOL	810.00		**IN** · money	THB	company	earnings	NPL's margin less the round-up it funded
CO.NONE.THB.EXP_ROUNDING	156.00		**IN** · cost	THB	company	rounding	the cost of rounding up (rounding_residual is negative)
CL.PROJECT.THB.PAYABLE		96,000.00	**OUT** · promise	THB	client	principal	the obligation is the rounded payout
CO.NONE.THB.EARN_GROSS		966.00	**OUT** · earned	THB	company	earnings	margin as priced

Rounding to the 1,000 applies to cash hand-overs only; a THB bank transfer goes out exact. The residual is signed: -156.00 THB here, so NPL funds the round-up from its margin and the client's claim is the whole 96,000.00. With a truncating option the sign is positive and the dropped fraction stays client money. Every rounding difference therefore has an owner (invariant 24).

9 · Pattern D — contingency routes and the bridge wallet

NPL's own wallets hold client money in three cases only (invariant 3): an approved reroute, a collection bridge for a network the partner lacks, and the own desk. The first two are shown here; the own desk is Pattern M.

T12 · Deal 3, Ali unavailable: 10,000 USDT collected into NPL's contingency wallet under an approved REROUTE source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWN.USDT.HELD	10,000.00		**IN** · money	USDT	client	principal	client money in custody, with deal, reason and age
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

Only an approved REROUTE, an own-desk leg or a bridge collection may debit CL.OWN.*.HELD (invariant 3). The custody view shows this balance until it is forwarded.

T13 · Forward to Jeton with recover dues: 10,000 USDT of client money credited at Jeton, 30.00 USDT of NPL's dues netted source: Reroute

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	Jeton credits the full client amount
CL.OWN.USDT.HELD		10,000.00	**OUT** · money	USDT	client	principal	custody ends
CO.OWN.USDT.WALLET	30.00		**IN** · money	USDT	company	earnings	NPL keeps 30 USDT in its wallet instead of sending it
CO.JETON.USDT.POOL		30.00	**OUT** · money	USDT	company	earnings	Jeton's debt to NPL falls by the same 30 USDT

NPL physically sends 9,970.00 USDT; Jeton treats it as 10,000 of client money because it owed NPL 30.00 USDT of earnings from earlier deals (partner_agreement_ref). The recovery is an asset swap, never income, and names the deal it recovers (invariant 15). **Nothing is withheld from the client amount:** NPL's margin is captured when Jeton converts (T14).

T14 · Conversion of deal 3 at Jeton (fee_on_market, 1.25 %): 10,000 USDT → 9,000.00 EUR at market, Jeton keeps 112.50 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	
CL.JETON.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	
CL.JETON.EUR.DUE	8,820.00		**IN** · money	EUR	client	principal	client net
CO.JETON.EUR.POOL	67.50		**IN** · money	EUR	company	earnings	NPL's net margin resting at Jeton
CO.NONE.EUR.EXP_PARTNER	112.50		**IN** · cost	EUR	company	partner_cost	Jeton's stated fee, deducted at transaction level
CL.PROJECT.EUR.PAYABLE		8,820.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		180.00	**OUT** · earned	EUR	company	earnings	gross margin before partner cost

Total cost to the client = partner cost 112.50 + NPL earnings 67.50 = 180.00 EUR. For a fee_in_rate partner (T2) the partner cost is inside the rate and no EXP_PARTNER line exists. Jeton rebates NPL 0.4 % of the amount (D27): a rebate accrual of 40.00 USDT posts with this conversion exactly as T33 shows for Aquanow, and is not repeated here.

Balances after the contingency route

Account	Ccy	Balance	Meaning
CL.OWN.USDT.HELD	USDT	0.00	custody ended
CL.JETON.EUR.DUE	EUR	8,820.00	client balance at Jeton
CO.JETON.EUR.POOL	EUR	67.50	NPL's net margin at Jeton
CO.OWN.USDT.WALLET	USDT	30.00	recovered dues in NPL's wallet
CO.NONE.EUR.EXP_PARTNER	EUR	112.50	Jeton's stated fees to date

T15 · Deal 4 (D40): the sender can only pay on ERC-20, which Ali does not accept; 10,000 USDT arrive on NPL's bridge wallet source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWN.USDT.HELD	10,000.00		**IN** · money	USDT	client	principal	client money on the bridge (OWN_WALLET.role collection_bridge), to be forwarded within custody_max_hours
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

T16 · Hop sent: the bridge forwards 10,000 USDT to Ali on TRC-20 (a SETTLEMENT_LINE to the partner_transit receiver, outcome sent) source: Settlement

No posting. No posting yet: a hop is not a delivery, so the promise stays and the money stays in the held jar until Ali credits it. The line carries deal_id (invariant 19) so the entitlement view keeps the leg's group.

T17 · Ali credits the hop: 10,000 USDT now sit at Ali as collected client money; network fee 4.20 USDT paid by NPL when one applies source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	the leg continues at its partner exactly as T1
CL.OWN.USDT.HELD		10,000.00	**OUT** · money	USDT	client	principal	custody ends
CO.NONE.USDT.EXP_NETWORK	4.20		**IN** · cost	USDT	company	network_fee	network fee of the hop, in USDT, NPL's cost (project network_fee_policy)
CO.OWN.USDT.WALLET		4.20	**OUT** · money	USDT	company	network_fee	paid from NPL's own USDT; when no network fee applies these two lines are absent

Nothing is earned or charged to the client on a hop. A network fee may or may not apply; when it does it is stated in USDT, and when it does not the fee is zero and no line posts.

10 · Pattern E — cancellations, reversals and voids

T18 · Erroneous posting: a collection of 2,000 USDT was booked against the wrong deal (as posted) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	2,000.00		**IN** · money	USDT	client	principal	booked in error
CL.PROJECT.USDT.PAYABLE		2,000.00	**OUT** · promise	USDT	client	principal	

T19 · Reversal of T18 (LEDGER_TRANSACTION.reversal_of_id = T18), then T18 is re-posted correctly source: Adjustment

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	2,000.00		**IN** · promise	USDT	client	principal	mirror image of T18
CL.ALI.USDT.COLLECTED		2,000.00	**OUT** · money	USDT	client	principal	

Nothing is edited or deleted: the wrong transaction stays, the reversal points at it, and the correct transaction follows (invariant 2). Any report at any past moment still adds up.

T20 · Collection of deal 5: 3,000 USDT arrive at Ali (the deal is voided afterwards) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT. COLLECTED	3,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT. PAYABLE		3,000.00	**OUT** · promise	USDT	client	principal	

T21 · Deal 5 voided after collection: refund to the sender in USDT, network fee 2.00 USDT charged back to the sender (project network_fee_policy) source: Adjustment

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT. PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	the obligation ends
CL.ALI.USDT. COLLECTED		2,998.00	**OUT** · money	USDT	client	principal	returned to the sender's wallet
CL.ALI.USDT. COLLECTED		2.00	**OUT** · money	USDT	client	network_fee	paid to the network; the sender bears it

If the policy, or a FEE_DECISION for this transaction, says NPL absorbs the fee, the last line becomes `Dr CO.NONE.USDT.EXP_NETWORK 2.00 / Cr CO.ALI.USDT.POOL 2.00` and the sender receives the full 3,000. Management approves the disposition of any post-money void.

11 · Pattern F — rejected payout and return leg

T22a · Collection of deal 6 at Ali: 2,000 USDT (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT. COLLECTED	2,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT. PAYABLE		2,000.00	**OUT** · promise	USDT	client	principal	

T22 · Conversion of deal 6 at Ali: 2,000 USDT → 1,544.40 GBP (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT. PAYABLE	2,000.00		**IN** · promise	USDT	client	principal	USDT leg
CL.ALI.USDT. COLLECTED		2,000.00	**OUT** · money	USDT	client	principal	USDT leg
CL.ALI.GBP.DUE	1,528.80		**IN** · money	GBP	client	principal	
CO.ALI.GBP.POOL	15.60		**IN** · money	GBP	company	earnings	
CL.PROJECT.GBP. PAYABLE		1,528.80	**OUT** · promise	GBP	client	principal	
CO.NONE.GBP. EARN_GROSS		15.60	**OUT** · earned	GBP	company	earnings	

T23 · Settlement of deal 6: 1,528.80 GBP released to the receiver's bank (the line names deal 6 in deal_id); the promise is discharged on release source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.GBP. PAYABLE	1,528.80		**IN** · promise	GBP	client	principal	sent: NPL no longer owes it unless it comes back
CL.ALI.GBP.DUE		1,528.80	**OUT** · money	GBP	client	principal	

T24 · The receiving bank rejects the transfer: 1,513.80 GBP come back to Ali, the bank kept 15.00 (SETTLEMENT_RETURN; line outcome = returned) source: SettlementReturn

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.GBP.DUE	1,513.80		**IN** · money	GBP	client	principal	back in the client balance at Ali
CL.PROJECT.GBP.PAYABLE		1,513.80	**OUT** · promise	GBP	client	principal	the promise is re-raised: NPL owes it again
CO.NONE.GBP.EXP_BANKFEE	15.00		**IN** · cost	GBP	company	bank_fee	bounce fee absorbed by NPL (treatment absorbed)
CO.ALI.GBP.POOL		15.00	**OUT** · money	GBP	company	bank_fee	funded from NPL's pool at Ali; if the GBP pool cannot cover it, it goes negative until T24a
CL.ALI.GBP.DUE	15.00		**IN** · money	GBP	client	bank_fee	NPL makes the client whole
CL.PROJECT.GBP.PAYABLE		15.00	**OUT** · promise	GBP	client	bank_fee	and the promise is whole again

The failed line is reversed before anything else happens (invariant 13), so the entitlement of the group deal 6 is attributed to is restored. The money can now stay in balance for a later payout or fund a return leg (T25).

T24a · Squaring NPL's GBP pool at Ali: with the client's next USDT conversion Ali also converts 19.43 USDT of NPL's own (its USDT pool there) into 15.00 GBP at 0.7722 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.USDT.FX_RESULT	19.43		**IN** · cost	USDT	company	earnings	currency sold: NPL's USDT leave its pool at Ali
CO.ALI.USDT.POOL		19.43	**OUT** · money	USDT	company	earnings	NPL's earnings kept in USDT at Ali (Pattern N) fund it
CO.ALI.GBP.POOL	15.00		**IN** · money	GBP	company	earnings	NPL's GBP pool is back to where it was before the bounce
CO.NONE.GBP.FX_RESULT		15.00	**OUT** · cost	GBP	company	earnings	currency bought: the GBP NPL received for them

NPL's own money converts at the partner's rate alongside the client's money; the partner sees one conversion. This is how a fee NPL absorbed in a currency where it had no pool is paid for: the pool went negative when the fee was funded, and the next conversion includes the equivalent in USDT. The FX_RESULT pair records that NPL sold USDT and bought GBP; valued in the reporting currency it is NPL's realised result on the exchange.

T25 · Return leg: 1,528.80 GBP converted back to 1,964.51 USDT at Ali (rate 1.285); markup waived by FEE_DECISION source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.GBP.PAYABLE	1,528.80		**IN** · promise	GBP	client	principal	GBP obligation ends
CL.ALI.GBP.DUE		1,528.80	**OUT** · money	GBP	client	principal	
CL.ALI.USDT.DUE	1,964.51		**IN** · money	USDT	client	principal	USDT owed to the sender's return receiver
CL.PROJECT.USDT.PAYABLE		1,964.51	**OUT** · promise	USDT	client	principal	

NPL's margin from T22 (15.60 GBP) stays in the pool: it was captured at the first conversion and is never at risk from a return. The round trip's rate loss falls on the client's funds. Had the FEE_DECISION applied a markup, part of the 1,964.51 USDT would have gone to CO.ALI.USDT.POOL and EARN_GROSS exactly as in T2. The return leg inherits its parent's group (D13); the payout to the sender's return receiver follows T23 and T8 with deal_id = the return leg (invariant 19).

12 · Pattern G — inter-group offsets

An offset is not an account. It arises only when a counterparty receiver of one group is paid from a leg attributed to another group (invariant 17): the settlement posts exactly as T7 and T8, and the `OFFSET` view, reading `GROUP_ENTITLEMENT`, shows one group owed and the other holding its flow. A project with a single group never has an offset; a cross-group payout needs Management approval (cross_group_payout) and tighter aging. No ledger line differs, which is the point: the money and the claim are right, only the attribution needs watching.

13 · Pattern H — balance conversion

T26 · Opening position: 5,000 SGD of converted client balance at Ali from earlier deals (summarised) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
`CL.ALI.SGD.DUE`	5,000.00		**IN** · money	SGD	client	principal	balance built up by earlier SGD deals
`CL.PROJECT.SGD.PAYABLE`		5,000.00	**OUT** · promise	SGD	client	principal	

T27 · Balance conversion at Ali: 5,000 SGD of client balance → 3,700.00 USDT at 0.74, no markup (serves Group B) source: BalanceConversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
`CL.PROJECT.SGD.PAYABLE`	5,000.00		**IN** · promise	SGD	client	principal	SGD obligation ends
`CL.ALI.SGD.DUE`		5,000.00	**OUT** · money	SGD	client	principal	SGD balance leaves
`CL.ALI.USDT.DUE`	3,700.00		**IN** · money	USDT	client	principal	USDT balance arrives
`CL.PROJECT.USDT.PAYABLE`		3,700.00	**OUT** · promise	USDT	client	principal	same obligation, new currency

No sender, receiver or collection: Finance requests, Management approves. ENTITLEMENT_REATTRIBUTION moves Group B's SGD entitlement into USDT; an excess over Group B's SGD entitlement is another group's flow and is booked as an approved offset. A banking-ready balance (USD, SGD) waits for conversion_due_date like a deal leg (D29); a markup, if any, is a FEE_DECISION and posts as in T2.

14 · Pattern I — overpayment and sender credit

T28 · Overpayment: a sender sends 10,050 USDT against a 10,000 USDT deal; 50 USDT becomes sender credit source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
`CL.ALI.USDT.COLLECTED`	10,050.00		**IN** · money	USDT	client	principal	all of it is client money at Ali
`CL.PROJECT.USDT.PAYABLE`		10,000.00	**OUT** · promise	USDT	client	principal	the deal
`CL.SENDER_A.USDT.CREDIT`		50.00	**OUT** · promise	USDT	client	principal	owed back to the sender (SENDER_CREDIT), or applied to the next deal

T29 · The credit is applied to the sender's next deal of 5,000 USDT: the sender pays 4,950 source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	4,950.00		**IN** · money	USDT	client	principal	what actually arrives
CL.SENDER_A.USDT.CREDIT	50.00		**IN** · promise	USDT	client	principal	the credit is consumed (state applied)
CL.PROJECT.USDT.PAYABLE		5,000.00	**OUT** · promise	USDT	client	principal	the full deal is collected

A refund instead of an application is a payout to the sender's return receiver: `Dr CL.SENDER_A.USDT.CREDIT 50 / Cr CL.ALI.USDT.COLLECTED 50` through an ordinary settlement line and confirmation. Sender credits are client money and are never netted against anything NPL is owed.

15 · Pattern J — markup shares and partner rebates

Both are accrued with the conversion they arise on (NPL, 10 October) and settled monthly against a statement. A share is a part of NPL's net markup owed to a party; a rebate is NPL's own income from a partner, kept apart from the margin and never in a share base (D26, D27).

T30 · Markup share, npl_pays (D26): an external markup-share party earns 25 % of NPL's earnings on deal 3 = 16.88 EUR, accrued with the conversion source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.EARN_GROSS	16.88		**IN** · earned	EUR	company	share	reduces NPL's margin
CO.PARTY_P.EUR.SHARE_PAYABLE		16.88	**OUT** · promise	EUR	company	share	owed to the party (MARKUP_SHARE_ACCRUAL direction payable)

Basis share_of_earnings. A fixed_pct party is priced into the sender rate instead and the posting has the same shape with the amount from the Calculation Specification. The accrual is posted in the same moment as T14, as its own transaction with source MarkupShare.

T31 · Month-end: NPL pays the party 16.88 EUR from its own wallet source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.PARTY_P.EUR.SHARE_PAYABLE	16.88		**IN** · promise	EUR	company	share	accrual state settled, paid_at set
CO.OWN.EUR.WALLET		16.88	**OUT** · money	EUR	company	share	or from the pool at a partner, when the partner pays the party on NPL's instruction

T32 · Raeen sub-account at Aquanow (party_retains, D26): 10,000 USDT → 8,928.00 EUR; sender pays 4 % on market, Aquanow charges 0.80 %; net markup 288.00 EUR shared 50 / 50 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	
CL.AQN.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	
CL.AQN.EUR.DUE	8,640.00		**IN** · money	EUR	client	principal	client net, in Raeen's sub-account at Aquanow, settled to Raeen's receivers as usual
CO.RAEEN.EUR.SHARE_RECEIVABLE	144.00		**IN** · money	EUR	company	share	NPL's remainder: Raeen holds the converted funds and owes it (direction receivable, period)
CO.NONE.EUR.EARN_GROSS	144.00		**IN** · earned	EUR	company	share	Raeen's share of the net markup; never NPL's money
CO.NONE.EUR.EXP_PARTNER	72.00		**IN** · cost	EUR	company	partner_cost	Aquanow's stated fee
CL.PROJECT.EUR.PAYABLE		8,640.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		360.00	**OUT** · earned	EUR	company	earnings	gross margin: the sender's 4 %

Share base = fee parts 360.00 − partner cost 72.00 = 288.00 EUR (invariant 25); the rebate below is not in the base. No POOL line: NPL's margin is not resting at the partner but with Raeen, so it is a receivable from Raeen instead. A second party with its own rule on the same leg adds one more pair of lines; the shares may not exceed 100 % of the base.

T33 · Partner rebate accrued with the same conversion (D27): Aquanow returns 0.3 % of the amount = 30.00 USDT source: Rebate

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.AQN.USDT. REBATE_RECEIVABLE	30.00		**IN** · money	USDT	company	rebate	expected from Aquanow (PARTNER_REBATE_ACCRUA state expected, period)
CO.NONE.USDT. EARN_REBATE		30.00	**OUT** · earned	USDT	company	rebate	NPL's alone: never in EARN_GROSS, never visible to Raeen (invariant 26)

The accrual names the NPLify deal it arises on; a partner statement is reconciled only against those deals. The accrual is expressed in the rule's payout_currency, the currency the partner pays in (Section 24).

T34 · Month-end: Aquanow's statement shows 29.50 USDT for the deal and pays it in USDT; the 0.50 USDT difference trues up the income source: Rebate

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.USDT.WALLET	29.50		**IN** · money	USDT	company	rebate	received (statement_ref on the accrual, state received)
CO.NONE.USDT. EARN_REBATE	0.50		**IN** · earned	USDT	company	rebate	true-up against the statement: the accrual's received_amount
CO.AQN.USDT. REBATE_RECEIVABLE		30.00	**OUT** · money	USDT	company	rebate	the expectation is cleared

A statement line that NPL cannot match to a deal is not posted: NPL only reconciles rebates on NPLify deals (D27).

T35 · Month-end: Raeen pays NPL its remainder 144.00 EUR source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.EUR.WALLET	144.00		**IN** · money	EUR	company	share	received
CO.RAEEN.EUR. SHARE_RECEIVABLE		144.00	**OUT** · money	EUR	company	share	accrual state settled, received_amount

Had Raeen paid in USDT, the receipt would land as `Dr CO.OWN.USDT.WALLET / Cr CO.NONE.USDT.FX_RESULT` and the receivable clear as `Dr CO.NONE.EUR.FX_RESULT / Cr CO.RAEEN.EUR.SHARE_RECEIVABLE` at the statement's agreed amounts: EUR sold, USDT bought, NPL's currency result (Section 3).

16 · Pattern K — variances, by kind

The sender's figures are write-once, so whenever the partner's actual conversion differs from the one the leg was priced on, the client's entitlement is unchanged and the difference is NPL's. Three of the four variance kinds have the same posting shape and differ only in the account; the fourth, fx timing, belongs to invoices and the own desk (Pattern M).

T36 · Partner calculates (calc_mode partner_calculates): Ali booked 3,000 USDT at 0.8905 instead of the 0.8910 version: expected 2,673.00 EUR, actual 2,671.50 EUR, variance 1.50 EUR

source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CO.ALI.EUR.POOL	25.50		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CONVERSION	1.50		**IN** · cost	EUR	company	variance	what the partner did worse than expected; beyond the project's tolerance an exception opens, the deal does not wait
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

The same account takes the basis variance of a reroute (the alternate partner's expected conversion on the net amount forwarded).

T37 · Cutoff timing (D29): 3,000 USDT locked at 0.8910 after Ali's 14:30 cutoff, converted next business day at 0.8890: expected 2,673.00 EUR, actual 2,667.00 EUR, variance 6.00 EUR

source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CO.ALI.EUR.POOL	21.00		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CUTOFF	6.00		**IN** · cost	EUR	company	cutoff_timing	rate stamped at lock vs rate used; NPL absorbs it
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

The leg stamps rate_locked_at, cutoff_at and conversion_due_date; Converting may not start before the due date (invariant 28). If the next day's rate would breach the minimum margin, the due date is pushed with conversion_deferred_reason and nothing posts until it runs.

T38 · Honoured expired rate (D45): a new leg priced on yesterday's 0.8910 by an approved override (honour_expired_rate); Ali converts today at 0.8860: expected 2,673.00 EUR, actual 2,658.00 EUR, variance 15.00 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CO.ALI.EUR.POOL	12.00		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_RATE_HONOUR	15.00		**IN** · cost	EUR	company	variance	the reused rate vs the day's rate: NPL's cost, chosen (invariant 39)
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

The leg stamps rate_reused_from_deal_id, rate_override_reason and rate_override_approved_by (Finance). A variance can also be a gain, in which case the line is a credit.

17 · Pattern L — loss events: booked to NPL first, recovered share by share

Money lost after it was verified or released: a frozen single-use account, cash found short and unrecoverable, a wrong-chain transfer (D32). The whole loss is NPL's expense at once and the client is made whole; the split Management approves is then recovered from each party by its own posting (NPL, 10 October).

T39 · Deal 7 collected: 10,000 USDT verified on a single-use local account at Ali (shown as T1)
source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

T40 · Loss event (D32): the account is frozen before conversion and the 10,000.00 USDT are lost; NPL books the whole loss and makes the client whole source: LossEvent

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.USDT.EXP_LOSS	10,000.00		**IN** · cost	USDT	company	loss	the whole loss is NPL's first (LOSS_EVENT state approved → posted)
CO.ALI.USDT.POOL		6,000.00	**OUT** · money	USDT	company	loss	NPL's money at Ali funds the make-good as far as it goes (LOSS_EVENT.pool_funded)
CO.OWN.USDT.WALLET		4,000.00	**OUT** · money	USDT	company	loss	the rest is transferred from NPL's own wallet to Ali (LOSS_EVENT.wallet_funded, funding_reference)
CL.ALI.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	the client money that was lost
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	loss	replaced by NPL's money: the deal proceeds, the collection part stays verified

A partial loss posts the same lines for the lost part only (LOSS_EVENT.amount ≤ the amount verified): if 4,000 of the 10,000 were lost, every line reads 4,000 and the other 6,000 convert normally. The client's claim (PAYABLE) is untouched and the deal converts and settles normally. When neither the pool nor an immediate transfer can fund the make-good, the unfunded part is posted as `Dr CL.PROJECT.USDT.MAKEGOOD / Cr CO.NONE.USDT.LOSS_PAYABLE` instead and the client balance at Ali is restored only when NPL's funds land there (then `Dr CL.ALI.USDT.COLLECTED / Cr CL.PROJECT.USDT.MAKEGOOD` and `Dr CO.NONE.USDT.LOSS_PAYABLE / Cr CO.OWN.USDT.WALLET`). A loss on the settlement side (money released and lost before the receiver had it) re-raises the promise and restores DUE from NPL's pool, from which a new settlement follows. A collection is moved to state reversed only when Management decides the deal is unwound instead; the make-good then becomes a refund to the sender.

T41a · Management approves the split (approve_loss_split): partner 4,000.00, sender 2,000.00, client 1,000.00, NPL 3,000.00 USDT — Ali's share agreed (LOSS_RECOVERY party partner, state agreed) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.LOSS_RECEIVABLE	4,000.00		**IN** · money	USDT	company	loss	Ali's agreed share, now owed to NPL
CO.NONE.USDT.EXP_LOSS		4,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls by Ali's share

One LOSS_RECOVERY row per share and one transaction per row (D46; rule 1): the three recoveries T41a–T41c are posted at the same moment, when Management approves the split, and each names its own row. The shares sum to the loss (invariant 31).

T41b · The sender's share agreed: 2,000.00 USDT (LOSS_RECOVERY party sender, state agreed) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.SENDER_A.USDT.LOSS_RECEIVABLE	2,000.00		**IN** · money	USDT	company	loss	the sender's agreed share, owed to NPL
CO.NONE.USDT.EXP_LOSS		2,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls by the sender's share

T41c · The client's share: 1,000.00 USDT borne by the client, received at once by partial reversal of the make-good (LOSS_RECOVERY party client, state received) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.POOL	1,000.00		**IN** · money	USDT	company	loss	NPL takes back the part of its make-good the client bears
CO.NONE.USDT.EXP_LOSS		1,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls to its own share, 3,000.00
CL.PROJECT.USDT.PAYABLE	1,000.00		**IN** · promise	USDT	client	loss	the client's claim falls by the share it bears
CL.ALI.USDT.COLLECTED		1,000.00	**OUT** · money	USDT	client	loss	and the restored balance with it

The client-share lines are the partial reversal of the make-good in T40, one of the ownership movements invariant 16 names; a client share is received the moment it is agreed, since no money changes hands.

T42 · Ali settles its share: 4,000.00 USDT credited to NPL's pool at Ali source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.POOL	4,000.00		**IN** · money	USDT	company	loss	received: the LOSS_RECOVERY row of T41a moves to state received
CO.ALI.USDT.LOSS_RECEIVABLE		4,000.00	**OUT** · money	USDT	company	loss	

T43 · The sender settles its share: 2,000.00 USDT paid to NPL's own wallet source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.USDT.WALLET	2,000.00		**IN** · money	USDT	company	loss	a payment to NPL, never a netting against a sender credit (client money); the LOSS_RECOVERY row of T41b moves to state received
CO.SENDER_A.USDT.LOSS_RECEIVABLE		2,000.00	**OUT** · money	USDT	company	loss	

Loss accounts after the recoveries

Account	Ccy	Balance	Meaning
CO.NONE.USDT.EXP_LOSS	USDT	3,000.00	NPL's own share of the loss, once every recovery is settled
CO.ALI.USDT.LOSS_RECEIVABLE	USDT	0.00	nothing outstanding from Ali
CO.SENDER_A.USDT.LOSS_RECEIVABLE	USDT	0.00	nothing outstanding from the sender

Net effect on NPL's money: the pool at Ali gave 6,000 and received 1,000 + 4,000 back (−1,000); the own wallet gave 4,000 and received 2,000 (−2,000); together −3,000, NPL's share. The client's claim on deal 7 is 9,000 USDT, backed by 9,000 at Ali.

18 · Pattern M — the reseller project (NPL-GR): two invoices, one remittance deal

NPL-GR, a business unit of NPL, buys from vendors at a preferential price and sells to its customers at a retail price (D23, D24). There are two invoices per deal: the vendor's invoice to NPL-GR, and NPL-GR's resale invoice to its customer, both in the vendor's currency. The difference is the game reseller revenue, and it must stay intact whatever happens afterwards. The customer pays the resale invoice through the remittance service with NPL's remittance markup on top; that markup pays for the remittance work, covers the partner's cost and absorbs the forex result of paying the vendor in instalments at different days' rates, and what is left is remittance earnings (NPL, 10 October).

The ledger therefore keeps three things apart. **The reseller books** are two postings per deal in the invoice currency, one per invoice, plus a claim on the remittance side. **The remittance deal** posts exactly like any client project, with NPL-GR as the client (holder PROJECT = the NPL-GR remittance project) and the resale invoice as the obligation the deal group pays. **The forex result** of each discharging line is a true-up between NPL-GR's USDT balance and NPL's pool, booked to VAR_FX_TIMING. The example uses NPL's own figures: a 9,000 EUR vendor invoice resold at 9,450 EUR, collected in THB, paid to the vendor in USDT in two instalments, the first at the group's invoice_rate (1.148369 USDT per EUR, 0.8708 the other way) and the second at 1.142505 (0.87527). Both payout routes are shown: Ali pays the first instalment; the balance hops to NPL's own desk, which pays the second and the margin line.

T44 · NPL-GR receives the vendor's invoice: 9,000.00 EUR at NPL's preferential price (INVOICE kind vendor) source: Invoice

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
`CO.NONE.EUR.RESELL_COST`	9,000.00		**IN** · cost	EUR	company	resale	NPL-GR's cost, in the invoice currency, when received
`CO.VENDOR_V.EUR.VENDOR_PAYABLE`		9,000.00	**OUT** · promise	EUR	company	resale	owed to the vendor; INVOICE_BALANCE shows the open part

These are NPL-GR's reseller books: two invoice postings per deal, both in the invoice currency, and nothing else. The multi-currency work is left to the remittance side below.

T45 · NPL-GR issues the resale invoice to its customer: 9,450.00 EUR at retail price (INVOICE kind resale); game reseller revenue 450.00 EUR is fixed here source: Invoice

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
`CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE`	9,450.00		**IN** · money	EUR	company	resale	owed by the customer
`CO.NONE.EUR.RESELL_REVENUE`		9,450.00	**OUT** · earned	EUR	company	resale	NPL-GR's revenue, in the invoice currency, when issued

RESELL_REVENUE − RESELL_COST = 450.00 EUR, and no later posting touches either account. The deal group that collects this invoice names it (DEAL_GROUP.invoice_id = the resale invoice, invoice_amount 9,450.00 EUR — the whole invoice in one collection here, an advance then the balance otherwise — and invoice_rate 1.148369 USDT per EUR, the base every instalment is reserved at). The resale invoice names the vendor invoice it marks up (vendor_invoice_id), which is how RESELLER_MARGIN pairs them.

T46 · The customer pays the resale invoice in THB cash to Ali: 359,746.78 THB = 9,450.00 EUR at the sender rate with the 2 % remittance markup, rounded up to 360,000.00 THB in cash (cash_rounding, unit 1,000) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.THB.COLLECTED	360,000.00		**IN** · money	THB	client	principal	client money at Ali — the client of this remittance project is NPL-GR
CL.PROJECT.THB.PAYABLE		360,000.00	**OUT** · promise	THB	client	principal	the remittance side owes it onward on NPL-GR's behalf
CO.PROJECT.EUR.REMIT_CLAIM	9,450.00		**IN** · money	EUR	company	resale	reseller books: the customer has paid, the remittance side now holds the value for NPL-GR
CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE		9,450.00	**OUT** · money	EUR	company	resale	the customer's debt is settled

The remittance side posts exactly like any client project (T1): the sender is a customer of NPL-GR's here, and NPL-GR is the client. The cash is counted and verified by Ali (D30) and the deal's amount is the rounded cash amount; the small surplus over the invoice equivalent is NPL-GR's and goes out with the margin line (T51). The two EUR lines are the reseller books recording that the customer paid; each owner's lines balance in their own currency.

T47 · Conversion at Ali (fee_in_rate, markup on market): 360,000.00 THB → 10,966.15 USDT at 32.83 (market 32.50); client net 10,859.73 USDT (= 10,852.09 for the 9,450.00 EUR at the group's invoice_rate 1.148369 USDT per EUR, plus the 7.63 USDT cash-rounding surplus), NPL's earnings 106.42 USDT source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.THB.PAYABLE	360,000.00		**IN** · promise	THB	client	principal	
CL.ALI.THB.COLLECTED		360,000.00	**OUT** · money	THB	client	principal	
CL.ALI.USDT.DUE	10,859.73		**IN** · money	USDT	client	principal	NPL-GR's entitlement in USDT, reserved for the vendor and for NPL-GR's margin
CO.ALI.USDT.POOL	106.42		**IN** · money	USDT	company	earnings	remittance earnings resting at Ali
CL.PROJECT.USDT.PAYABLE		10,859.73	**OUT** · promise	USDT	client	principal	
CO.NONE.USDT.EARN_GROSS		106.42	**OUT** · earned	USDT	company	earnings	0.96 % of the market value of the THB: the 2 % markup on the target (217.19) less Ali's 1 % inside its rate (110.77)

The markup is applied on the market rate because the price is built from the EUR invoice; Ali's agent rate is used for the THB → USDT conversion and only decides whether 2 % leaves enough: the fee structure's minimum margin is checked against the spread between market and agent rate (FEE_STRUCTURE mode variable, min_margin_pct, floor and cap), and when Ali's rate leaves less, the markup rises within the cap or Finance asks for a better rate. The markup pays for the remittance service and carries the rate risk of the instalments, so the 450.00 EUR reseller revenue is never diluted. Had the customer paid in USDT, the pair would be USDT → USDT at rate 1 (D33) and the same lines would post in USDT.

T48 · Instalment 1 paid by Ali: 3,445.11 USDT to the vendor, discharging 3,000.00 EUR of the vendor invoice at discharge_rate 1.148369 USDT per EUR (release; the vendor's confirmation posts nothing more)

source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,445.11		**IN** · promise	USDT	client	principal	released by Ali on NPL's instruction: the promise is discharged
CL.ALI.USDT.DUE		3,445.11	**OUT** · money	USDT	client	principal	
CO.VENDOR_V.EUR.VENDOR_PAYABLE	3,000.00		**IN** · promise	EUR	company	resale	reseller books: the vendor is owed 3,000 EUR less (vendor INVOICE_BALANCE 6,000, counted over released lines)
CO.PROJECT.EUR.REMIT_CLAIM		3,000.00	**OUT** · money	EUR	company	resale	the remittance side holds 3,000 EUR less for NPL-GR

The line names the vendor invoice (SETTLEMENT_LINE.invoice_id, obligation_discharged 3,000.00 EUR, discharge_rate 0.8708). Paid on the pricing day, so the USDT reserved for these 3,000 EUR (3,445.11) equals the USDT paid: no forex line.

T49 · Variant route: the remaining 7,414.62 USDT of NPL-GR's balance hop from Ali to NPL's own desk (LT Sub at Aquanow; a SETTLEMENT_LINE to the partner_transit receiver, posted when the own desk credits)

source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWNDESK.USDT.DUE	7,414.62		**IN** · money	USDT	client	principal	credited at the own desk: an ordinary partner-coded account with the own-desk configuration as holder (D23), reported by the custody view
CL.ALI.USDT.DUE		7,414.62	**OUT** · money	USDT	client	principal	left Ali; a hop is not a delivery, so the promise is untouched

Whether Ali or the own desk pays the vendor depends on the vendor; the postings that follow are identical apart from the holder. Nothing is earned or charged on the hop; a network fee NPL bears posts as in T17.

T50 · Instalment 2 paid by the own desk four days later: 6,855.03 USDT discharge the remaining 6,000.00 EUR at discharge_rate 1.142505 USDT per EUR; 6,890.22 USDT had been reserved at the group's invoice_rate 1.148369, so 35.19 USDT are a forex gain for the remittance side source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	6,855.03		**IN** · promise	USDT	client	principal	released: the promise is discharged
CL.OWNDESK.USDT.DUE		6,855.03	**OUT** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE	35.19		**IN** · promise	USDT	client	fx_timing	forex true-up: the 6,000 EUR cost fewer USDT than reserved, the surplus is not NPL-GR's
CL.OWNDESK.USDT.DUE		35.19	**OUT** · money	USDT	client	fx_timing	it leaves NPL-GR's balance
CO.OWNDESK.USDT.POOL	35.19		**IN** · money	USDT	company	fx_timing	and becomes NPL's, resting at the own desk
CO.NONE.USDT.VAR_FX_TIMING		35.19	**OUT** · cost	USDT	company	fx_timing	a gain; with the rate the other way the line is a debit and the pool tops NPL-GR's balance up
CO.VENDOR_V.EUR.VENDOR_PAYABLE	6,000.00		**IN** · promise	EUR	company	resale	reseller books: the vendor invoice is settled (INVOICE_BALANCE 0)
CO.PROJECT.EUR.REMIT_CLAIM		6,000.00	**OUT** · money	EUR	company	resale	

The forex true-up is the one place where the remittance side's USDT obligation to NPL-GR is re-measured: NPL-GR is entitled to EUR value, the markup absorbs the rate movement either way (NPL's 1.3 % less a 0.1 % forex loss gives 1.2 % in NPL's own example), and the reseller books never see it.

T51 · The margin line: 450.00 EUR of the resale invoice (514.13 USDT at 1.142505 USDT per EUR) plus the 7.63 USDT cash-rounding surplus are paid to NPL-GR's own wallet, 521.76 USDT in all; 516.77 USDT had been reserved for the margin, 2.64 USDT forex gain source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	521.76		**IN** · promise	USDT	client	principal	released to NPL-GR's wallet (a receiver of the project); the balance is empty. The resale invoice's own balance went to zero at T46, when the group that collects its whole amount was paid
CL.OWNDESK.USDT.DUE		521.76	**OUT** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE	2.64		**IN** · promise	USDT	client	fx_timing	forex true-up on the margin line
CL.OWNDESK.USDT.DUE		2.64	**OUT** · money	USDT	client	fx_timing	
CO.OWNDESK.USDT.POOL	2.64		**IN** · money	USDT	company	fx_timing	
CO.NONE.USDT.VAR_FX_TIMING		2.64	**OUT** · cost	USDT	company	fx_timing	
CO.OWN.USDT.WALLET	521.76		**IN** · money	USDT	company	resale	reseller books: NPL-GR's margin arrives, in USDT
CO.NONE.USDT.FX_RESULT		514.13	**OUT** · cost	USDT	company	resale	currency bought: the USDT received for the 450 EUR claim at the day's rate
CO.NONE.USDT.RESELL_REVENUE		7.63	**OUT** · earned	USDT	company	rounding	the cash-rounding surplus is NPL-GR's revenue in the currency received, never a forex result
CO.NONE.EUR.FX_RESULT	450.00		**IN** · cost	EUR	company	resale	currency sold: the EUR claim given up
CO.PROJECT.EUR.REMIT_CLAIM		450.00	**OUT** · money	EUR	company	resale	nothing is held for NPL-GR any more

NPL-GR may instead leave its margin in the project balance to fund the next vendor payment (settlement_policy hold_allowed): then nothing in T51 posts, REMIT_CLAIM keeps 450.00 EUR, the remittance side keeps the matching USDT payable, and the forex true-up of the margin waits for the line that pays it. The 7.63 USDT surplus comes from the cash having been rounded up at collection; it is NPL-GR's, has no EUR invoice behind it, and simply arrives in the wallet. FX_RESULT records 450.00 EUR sold against 514.13 USDT bought at 1.142505 USDT per EUR, NPL's realised currency result, valued through REPORTING_VALUE.

Balances after the resale invoice is fully discharged

Account	Ccy	Balance	Meaning
CO.NONE.EUR.RESELL_REVENUE	EUR	9,450.00	resale invoice, untouched since issue
CO.NONE.EUR.RESELL_COST	EUR	9,000.00	vendor invoice, untouched since receipt
CO.VENDOR_V.EUR.VENDOR_PAYABLE	EUR	0.00	vendor settled
CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE	EUR	0.00	customer paid
CO.PROJECT.EUR.REMIT_CLAIM	EUR	0.00	nothing held for NPL-GR
CL.OWNDESK.USDT.DUE	USDT	0.00	NPL-GR's balance at the own desk, fully paid out
CO.OWNDESK.USDT.POOL	USDT	37.83	the forex gains, resting at the own desk (the remittance earnings rest in NPL's pool at Ali, which this example's loss and squaring patterns also touched)

Account	Ccy	Balance	Meaning
CO.NONE.USDT.EARN_GROSS	USDT	106.42	remittance earnings on the deal
CO.NONE.USDT.VAR_FX_TIMING	USDT	-37.83	forex result (negative = gain)
CO.NONE.EUR.FX_RESULT	EUR	450.00	NPL-GR's margin claim settled in USDT: the EUR leg
CO.NONE.USDT.FX_RESULT	USDT	-494.70	the USDT leg (negative = credit); includes the 19.43 USDT of the pool squaring in T24a

Reseller revenue 450.00 EUR, intact. Remittance margin on the deal 144.25 USDT = earnings 106.42 + forex 37.83, which is 1.30 % of the market value of the THB collected; the 7.63 USDT cash-rounding surplus is NPL-GR's, not the remittance side's. Bank fees on a vendor payment, a bounced vendor transfer, a loss event and a rebate from Ali post exactly as in the other patterns, with NPL-GR as the client. A loss on NPL-GR's funds is made good like any client's (Pattern L) and the client share, if any, is NPL-GR's.

19 · Pattern N — earnings kept in the unconverted balance and offset between partners

Partners convert everything they receive, so NPL's margin is converted along with the client's money and rests at the partner in the payout currency, invisible to the partner (EARNINGS_RECEIVABLE, Pattern A). NPL's practice (10 October) is to take its earnings out of collections that are not yet converted instead, and to offset between partners: where Jeton has converted a large collection in full, and Ali holds a collection that converts tomorrow, NPL leaves in Ali's USDT balance the USDT equivalent of its margin at Jeton and of its margin on the Ali deal, and converts only the rest. Jeton's EUR balance is then entirely the client's; at Ali the USDT balance is NPL's and the EUR balance the client's.

In the ledger this is two things. **Retention** is a conversion that keeps the deal's own margin in the collection currency (CONVERSION.residual_in), so EARN_GROSS is recognised in USDT rather than EUR. **The offset** is an exchange at the day's rate: NPL's EUR at one partner becomes client money there, the same value of the client's unconverted USDT at another partner becomes NPL's, and the client's claim is re-denominated accordingly. For the client it is a conversion NPL performed out of its own earnings; for NPL it is a currency swap, recorded in FX_RESULT. Neither moves money physically, and neither changes any client's entitlement.

Figure	Jeton deal (300,000 USDT)	Ali deal (100,000 USDT)	How
Market rate	0.9000	0.9000	EUR per USDT, the rate the sender is priced on
Sender rate	0.8874	0.8874	market × (1 – 1.40 % sender pays)
Gross out	266,220.00	88,740.00	amount × sender rate
Receiver pays	1,620.00	540.00	0.60 % of amount × market
Client net → DUE and PAYABLE	264,600.00	88,200.00	amount × market × (1 – 2.00 %): the client's entitlement, write-once
Partner rate used	0.9000	0.8910	Jeton converts at market and charges separately; Ali's 1 % is inside its rate
Converted (gross)	270,000.00	89,100.00	amount × partner rate
Partner cost → EXP_PARTNER	3,375.00	0.00 (900.00 inside the rate)	Jeton 1.25 % of converted, booked; Ali's is never posted
Delivered by the partner	266,625.00	89,100.00	converted – stated partner cost
EARN_GROSS posted	5,400.00	900.00	converted – client net: for Jeton the full fee parts (its cost is booked separately as EXP_PARTNER); for Ali the fee parts 1,800.00 less its hidden cost 900.00
NPL's margin → POOL	2,025.00 EUR	900.00 EUR	delivered – client net: what rests at the partner for NPL
Offset, USDT equivalent	2,272.73 USDT		Jeton margin ÷ Ali's rate 0.891: taken from the client's USDT at Ali, given back as EUR at Jeton (FX_RESULT: EUR out, USDT in)

Figure	Jeton deal (300,000 USDT)	Ali deal (100,000 USDT)	How
Retained, USDT equivalent		1,010.10 USDT	Ali margin ÷ Ali's rate: left unconverted, EARN_GROSS in USDT
USDT Ali actually converts		96,717.17 USDT	100,000 – offset – retained
EUR delivered at Ali → DUE		86,175.00 EUR	= Ali client net – the 2,025.00 already delivered at Jeton
NPL's USDT pool at Ali		3,282.83 USDT	offset + retained: both margins, now in USDT at Ali

T52a · Collection at Jeton: 300,000 USDT (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.USDT.COLLECTED	300,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		300,000.00	**OUT** · promise	USDT	client	principal	

T52 · Jeton converts the 300,000 USDT in full → 266,625.00 EUR; NPL's margin 2,025.00 EUR now sits at Jeton in EUR (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	300,000.00		**IN** · promise	USDT	client	principal	the USDT promise is discharged by the conversion
CL.JETON.USDT.COLLECTED		300,000.00	**OUT** · money	USDT	client	principal	
CL.JETON.EUR.DUE	264,600.00		**IN** · money	EUR	client	principal	client net (dust ignored here)
CO.JETON.EUR.POOL	2,025.00		**IN** · money	EUR	company	earnings	NPL's margin, in EUR, inside Jeton's balance; Jeton does not know it is there
CO.NONE.EUR.EXP_PARTNER	3,375.00		**IN** · cost	EUR	company	partner_cost	Jeton's stated 1.25 %
CL.PROJECT.EUR.PAYABLE		264,600.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		5,400.00	**OUT** · earned	EUR	company	earnings	

The normal case: the partner converts everything it received, so NPL's margin is converted with it and rests at that partner (EARNINGS_RECEIVABLE). Partners never see NPL's margin; what they hold is one balance.

T53 · Meanwhile 100,000 USDT arrive at Ali for another deal, scheduled to convert the next business day (D29) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	100,000.00		**IN** · money	USDT	client	principal	unconverted at Ali
CL.PROJECT.USDT.PAYABLE		100,000.00	**OUT** · promise	USDT	client	principal	

T54 · Earnings offset between partners (EARNINGS_OFFSET): NPL's 2,025.00 EUR at Jeton become client money there; 2,272.73 USDT of the client's unconverted balance at Ali (= 2,025.00 EUR at Ali's rate 0.891) become NPL's source: EarningsOffset

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.EUR.DUE	2,025.00		**IN** · money	EUR	client	earnings	Jeton's EUR balance is now entirely client money
CL.PROJECT.EUR.PAYABLE		2,025.00	**OUT** · promise	EUR	client	earnings	the client's EUR claim grows by the same amount
CL.PROJECT.USDT.PAYABLE	2,272.73		**IN** · promise	USDT	client	earnings	and its USDT claim falls: for the client this is a conversion of USDT at Ali into EUR at Jeton, at the day's rate
CL.ALI.USDT.COLLECTED		2,272.73	**OUT** · money	USDT	client	earnings	that much USDT at Ali is no longer the client's
CO.ALI.USDT.POOL	2,272.73		**IN** · money	USDT	company	earnings	NPL's margin now rests at Ali, in USDT, inside the unconverted balance
CO.NONE.USDT.FX_RESULT		2,272.73	**OUT** · cost	USDT	company	earnings	currency bought: the USDT NPL took at Ali
CO.NONE.EUR.FX_RESULT	2,025.00		**IN** · cost	EUR	company	earnings	currency sold: the EUR NPL gave up at Jeton
CO.JETON.EUR.POOL		2,025.00	**OUT** · money	EUR	company	earnings	nothing of NPL's rests at Jeton any more; the EARNINGS_RECEIVABLE rows at Jeton are recovered by this offset

Finance initiates, Management approves, and the offset names the deals it recovers and the rate it used. Nothing moves physically. For the client the effect is neutral: the same EUR value, part of it now at Jeton instead of Ali. For NPL the margin earned in EUR is now held in USDT: FX_RESULT records EUR sold and USDT bought at the day's rate, NPL's realised exchange, and nothing needs closing later.

T55 · Ali converts the next day, less NPL's margins: 96,717.17 USDT → 86,175.00 EUR; the Ali deal's own margin 900.00 EUR is kept as 1,010.10 USDT instead (CONVERSION.residual_in) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	97,727.27		**IN** · promise	USDT	client	principal	the client's remaining USDT claim at Ali
CL.ALI.USDT.COLLECTED		96,717.17	**OUT** · money	USDT	client	principal	what Ali actually converts
CL.ALI.USDT.COLLECTED		1,010.10	**OUT** · money	USDT	client	earnings	NPL's margin on this deal, left unconverted
CO.ALI.USDT.POOL	1,010.10		**IN** · money	USDT	company	earnings	and recognised in USDT
CO.NONE.USDT.EARN_GROSS		1,010.10	**OUT** · earned	USDT	company	earnings	= 900.00 EUR at 0.891: the margin is earned in the collection currency
CL.ALI.EUR.DUE	86,175.00		**IN** · money	EUR	client	principal	client net less the part already delivered at Jeton
CL.PROJECT.EUR.PAYABLE		86,175.00	**OUT** · promise	EUR	client	principal	the client's EUR entitlement on this deal (write-once) less what it already holds at Jeton

The client's entitlement is unchanged: 88,200.00 EUR as priced, of which 2,025.00 already sit at Jeton and 86,175.00 now at Ali. Ali's balance after this is 3,282.83 USDT (all NPL's) plus 86,175.00 EUR (all the client's). The pool identity still holds: NPL's USDT at Ali is inside the one USDT balance Ali reports.

Change in balances over Pattern N

Account	Ccy	Change	Meaning
CL.JETON.EUR.DUE	EUR	+266,625.00	client money at Jeton: the deal's client net plus NPL's former margin there
CO.JETON.EUR.POOL	EUR	+0.00	nothing of NPL's rests at Jeton
CL.ALI.USDT.COLLECTED	USDT	+0.00	nothing of the client's unconverted at Ali
CO.ALI.USDT.POOL	USDT	+3,282.83	NPL's margins from both partners, in USDT at Ali
CL.ALI.EUR.DUE	EUR	+86,175.00	client money at Ali
CL.PROJECT.EUR.PAYABLE	EUR	+352,800.00	the client's EUR claim: both deals' client nets in full
CL.PROJECT.USDT.PAYABLE	USDT	+0.00	no USDT claim left
CO.NONE.EUR.EARN_GROSS	EUR	+5,400.00	Jeton deal's margin, earned in EUR
CO.NONE.USDT.EARN_GROSS	USDT	+1,010.10	Ali deal's margin, earned in USDT
CO.NONE.EUR.FX_RESULT	EUR	+2,025.00	the EUR NPL gave up at Jeton
CO.NONE.USDT.FX_RESULT	USDT	−2,272.73	the USDT it took at Ali (negative = credit)

The offset is approximate only in the sense that NPL chooses the rate (the partner's rate of the day in the example); the ledger is exact at that rate. The same mechanism recovers EARNINGS_RECEIVABLE rows without waiting for the partner's fee cycle, and it is the everyday counterpart of the netting on a reroute (T13). Where NPL later converts its USDT pool at Ali into EUR, that is another exchange of company money, posted to FX_RESULT in the same way; the reporting-currency value of FX_RESULT over the period is NPL's currency result.

20 · Identities and controls

Identity	Statement
Client assets = client liabilities, per currency	Σ COLLECTED + Σ HELD + Σ DUE + Σ SHORTFALL + MAKEGOOD = PAYABLE + Σ CREDIT. Holds after every transaction because every transaction balances per owner as well as per currency. Money released to a receiver is in neither side: it is out of NPL's hands and the promise for it is discharged.
PAYABLE = Σ GROUP_ENTITLEMENT	By construction (invariant 7, D46 amendment): every PAYABLE line carries a receiver_group_id — the leg's group, the paying leg's group on a release, return or short confirmation (deal_id, invariant 19), the served group on a balance conversion, the deal's group on a loss share or an offset — and the view sums those lines.
PROJECT_BALANCE(partner) = CL.<PARTNER>.<CCY>.DUE	The view is the account balance (D46 amendment: it was a sum over conversions' actual_out, which includes NPL's margin and had no term for top-ups, make-goods, offsets or hops); the pending converted value of a locked, not yet converted leg (D29) is shown by the view and is not posted.
Pool per partner = the partner's statement	Per partner and currency, client balances plus NPL's pool equal the one balance the partner reports; a partner never sees the split. Σ EARNINGS_RECEIVABLE outstanding ≤ POOL, and an EARNINGS_OFFSET recovers receivables at one partner into a pool at another.
Rebates and shares agree to statements	Σ REBATE_RECEIVABLE = Σ PARTNER_REBATE_ACCRUAL expected – received, per partner; Σ SHARE_RECEIVABLE / SHARE_PAYABLE = Σ MARKUP_SHARE_ACCRUAL not yet settled, per party.
Loss accounting closes	Per LOSS_EVENT: EXP_LOSS net of recoveries = npl_share once every share is settled; Σ LOSS_RECEIVABLE = agreed shares not yet received.
Exposure ≤ ceiling	Σ DUE at a partner, plus its lines sent and not yet confirmed, is within the project's exposure ceiling; Σ PROJECT_BALANCE across a client's projects, as REPORTING_VALUE, within CLIENT.exposure_ceiling (invariant 35).
Own wallets hold client money only in three cases	CL.OWN.<CCY>.HELD is zero except for rows the custody view can name: an open reroute, a bridge collection within custody_max_hours, an own-desk leg (invariant 3).
Every released line is accounted for	Per SETTLEMENT_LINE: confirmed + shortfall re-raised + returned + fees booked = the amount released; a line never keeps outcome sent beyond the rail's expected time without an exception.
A pool may be negative only between a top-up and the next conversion	CO.<PARTNER>.<CCY>.POOL below zero means the partner fronted a fee NPL absorbed in a currency where NPL held nothing; it is squared at the next conversion by converting NPL's own USDT alongside the client's (T24a).
Reseller revenue is intact	Per resale invoice: RESELL_REVENUE − RESELL_COST is set when the two invoices are posted and no later transaction touches either account in the invoice currency (a cash-rounding surplus is extra revenue, posted to RESELL_REVENUE in the currency received, never a forex result). Σ VENDOR_PAYABLE = Σ vendor-invoice balances over released lines; Σ RESALE_RECEIVABLE = Σ resale-invoice balances over collected groups (DEAL_GROUP.invoice_amount); REMIT_CLAIM = what the remittance side holds for NPL-GR, in EUR.
Remittance margin per deal	EARN_GROSS − EXP_PARTNER ± VAR_FX_TIMING − fees on the deal's lines, in the converted currency; the forex result of the instalments lands here and nowhere else.

Identity	Statement
Every figure shown is a sum	No screen stores a balance; all are sums over postings or deals (invariant 10). A converted total is a REPORTING_VALUE: an estimate at the latest platform-pair rate, never posted (invariant 38).

Month-end, Finance runs the identities above and reconciles three statements: each partner's statement of what it holds for NPL (POOL), each partner's rebate statement (REBATE_RECEIVABLE, NPLify deals only), and each markup-share party's statement (SHARE_RECEIVABLE / SHARE_PAYABLE). A difference is a true-up posting that names the statement line, never an edit.

21 · Reports the ledger answers

- **Where the money is.** Client money by partner, sent and not yet confirmed (read from the payout lines, not from an account), in custody (CUSTODY view: reroutes, bridge, own desk), shortfalls and credits; NPL's money by partner (POOL) and wallet (WALLET), receivables and payables by counterparty.
- **Client position.** PROJECT_BALANCE per partner and GROUP_ENTITLEMENT per group, rolled up per client (CLIENT_POSITION) with the period's EARN_GROSS and EARN_REBATE; compared with the client's exposure ceiling as a REPORTING_VALUE.
- **Profit and loss, per currency.** EARN_GROSS (net of shares) + EARN_REBATE + RESELL_REVENUE − RESELL_COST − EXP_PARTNER − EXP_BANKFEE − EXP_NETWORK − EXP_ROUNDING − EXP_LOSS ± the four VAR_ accounts, by project, partner, pair and month, from the cost-component and source tags. A total across currencies is a REPORTING_VALUE with its rate and time.
- **Partner commissions.** Per partner and period: margin captured there, dues outstanding (EARNINGS_RECEIVABLE), rebates expected, reconciled and received (statement_ref).
- **Markup shares.** Per party and period: base, share, direction, settled and open amounts; a party never sees a rebate.
- **Variance and loss.** The four variance kinds by partner and month, each line traceable to its leg; loss events with their shares and recovery state.
- **Reseller and remittance margins, kept apart, and NPL's profit.** Per resale invoice: revenue, cost and game reseller revenue in EUR, intact (RESELLER_MARGIN). Per NPL-GR deal: remittance earnings, partner cost, forex result and fees in the converted currency (REMITTANCE_MARGIN). NPL's profit (NPL_PROFIT) is remittance earnings plus reseller earnings, each callable on its own and together, per currency and as one figure through REPORTING_VALUE.

22 · Transaction sources and what they post

Source event	Posts to	Owner tags	Cost components
Invoice	RESELL_COST and VENDOR_PAYABLE (vendor invoice); RESALE_RECEIVABLE and RESELL_REVENUE (resale invoice)	company	resale
Collection	COLLECTED or HELD, PAYABLE, CREDIT; in NPL-GR also REMIT_CLAIM and RESALE_RECEIVABLE	client (and company)	principal, resale
Conversion (deal leg)	COLLECTED / PAYABLE in the in-currency; DUE, PAYABLE, POOL, EARN_GROSS, EXP_PARTNER, EXP_ROUNDING, SHARE_RECEIVABLE, a VAR_ account in the out-currency	client and company	principal, rounding, earnings, partner_cost, share, variance, cutoff_timing

Source event	Posts to	Owner tags	Cost components
BalanceConversion	DUE and PAYABLE in both currencies; POOL and EARN_GROSS if a markup is decided	client (and company)	principal, rounding, earnings
Settlement	PAYABLE and DUE on release (the promise is discharged); EXP_BANKFEE and POOL when a fee is absorbed up front. A hop posts nothing on release. Each line names the leg it settles (deal_id) and, in NPL-GR, the invoice it discharges	client (and company)	principal, bank_fee
SettlementReturn	DUE and PAYABLE (the promise is re-raised); EXP_BANKFEE and POOL for the bounce fee	client and company	principal, bank_fee
Confirmation	Nothing on a full delivery; SHORTFALL and PAYABLE on a short one; a hop credited: COLLECTED or DUE at the partner from HELD or DUE, with EXP_NETWORK and WALLET for a network fee	client (and company)	principal, bank_fee, network_fee
BankFeeEvent	EXP_BANKFEE, POOL, DUE, SHORTFALL	company and client	bank_fee
Reroute	HELD, COLLECTED at the alternate partner; WALLET and POOL for recovered dues; VAR_CONVERSION for basis variance	client and company	principal, earnings, variance
EarningsOffset	DUE and POOL at the partner holding NPL's margin; COLLECTED and POOL at the partner holding unconverted client money; PAYABLE in both currencies; FX_RESULT	client and company	earnings
MarkupShare	EARN_GROSS, SHARE_PAYABLE or SHARE_RECEIVABLE; WALLET or POOL on settlement	company	share
Rebate	REBATE_RECEIVABLE, EARN_REBATE; WALLET on receipt	company	rebate
LossEvent	EXP_LOSS, POOL or WALLET (or LOSS_PAYABLE), COLLECTED / DUE (or MAKEGOOD), PAYABLE re-raised on a settlement-side loss	company and client	loss, principal
LossRecovery	LOSS_RECEIVABLE against EXP_LOSS when a partner or sender share is agreed; POOL or WALLET against LOSS_RECEIVABLE when it is received; POOL against EXP_LOSS with PAYABLE and COLLECTED / DUE for the client's share, a partial reversal of the make-good	company (and client)	loss
Adjustment	any, always as a reversal plus a correct re-posting, a Management-approved disposition, or a company exchange of one currency for another through FX_RESULT	as reversed	as reversed

23 · Who sees what, and who may post

Rights are rows, not text (D44): `ROLE` holds the three roles of today, operations, finance and management, and `ROLE_RIGHT` says per controlled action which role may initiate and which may approve, with the role added on escalation; the approver is never the initiator (invariant 9).

- **Operations** (sees_economics false) sees operational fields, sender-facing prices and partner rates as entered, but never a ledger balance, a margin, a pool, a share, a rebate, a cost breakdown or a variance, on any screen, export, message or report.
- **Finance** sees all postings and balances, owns reconciliation and the month-end true-ups, initiates adjustments, balance conversions and dues recovery, recommends covered bank fees and fee caps

(FEE_DECISION.recommended_by), approves honoured expired rates and quote-validity extensions, and overrides a screening reject.

- **Management** approves the actions the ledger patterns depend on: approve_loss_split, balance_conversion, recover_dues, earnings_offset (requested by Finance), cross_group_payout, cover_bank_fee, raise_fee_cap, enable_own_desk, settlements above limit and past the cutoff, own-wallet outbound transfers, and post-money voids. Every controlled action is audited with actor, time, reason and record version.

24 · What this design put into the ERD (D46 in v5.7, amended as v5.8, propagated in v5.9)

The patterns above required the following additions, agreed with NPL on 10–11 October and cut into ERD Draft v5.7 as decision D46. Recovery postings use the source LossRecovery; the reseller books the source Invoice; the offset the source EarningsOffset. An offset records the to-partner's rate of the day (NPL's choice over the market rate), and the reseller books are postings rather than invoice records with views, so that NPL's profit (NPL_PROFIT) can be asked for as remittance earnings, reseller earnings, or both. An independent re-derivation of every calculation from the ERD and this document, run on 11 October, found five gaps between the model's definitions and these postings; they were applied the same evening as an amendment of D46, cut as ERD v5.8, and are marked below. A cross-document review the same night found no new gap in the design but a number of passages — in the ERD, in this document and in the Companion — that still described the pre-D46 ledger; v5.9 corrects them, as noted below.

1. **Variance by kind.** `LEDGER_ACCOUNT.purpose`: VARIANCE replaced by VAR_CONVERSION, VAR_CUTOFF, VAR_RATE_HONOUR, VAR_FX_TIMING. Invariants 28 and 39 and decisions D24, D29, D45 then name the account they post to (the three decision texts and the DEAL rate-reuse note were brought into line in v5.9).
2. **Loss first, recover later.** Purposes LOSS_RECEIVABLE (holder Partner or Sender), and the pair MAKEGOOD (client asset, holder Project) / LOSS_PAYABLE (company liability) for an unfunded make-good. A table `LOSS_RECOVERY` (loss_event_id, party kind sender / partner / client, amount, state agreed / received, received_at, reference, approved_by) as the source record of each recovery posting, under source LossRecovery; each state change (agreed, received) posts one transaction naming the row, and T41 is shown as one transaction per share (v5.9). `LOSS_EVENT` records how the make-good was funded (pool_funded, wallet_funded, funding_reference; v5.9). `COLLECTION.state` reversed is used only when the deal is unwound; invariant 31 says so since v5.9. Invariant 16 restated (amendment) as two lists — client → company: margin recognition, the client side of an offset, a forex true-up gain, the client's loss share; company → client: absorbed or covered fees, round-ups, top-ups, make-wholes, make-goods, the company side of an offset, a forex true-up loss.
3. **Accrual timing.** `MARKUP_SHARE_ACCRUAL` and `PARTNER_REBATE_ACCRUAL` are created and posted by the conversion that gives rise to them (invariant 25 and 26 restated to say so); `PARTNER_REBATE_RULE.payout_currency` names the currency the partner pays in, in which the accrual is expressed. *Amendment:* basis keeps pct_of_amount only; pct_of_partner_fee, retired with the D27 amendment, is dropped.
4. **Reseller project.** `INVOICE.kind` vendor / resale, with `receiver_id` for the vendor and `sender_id` for the customer of a resale invoice; `DEAL_GROUP.invoice_id` names the resale invoice the deal collects; `SETTLEMENT_LINE.invoice_id` names the invoice a line discharges (vendor invoice for a vendor line, resale invoice for the margin line), with `obligation_discharged` and `discharge_rate` (payout currency per invoice currency) as today; purposes RESELL_REVENUE, RESELL_COST, VENDOR_PAYABLE, RESALE_RECEIVABLE, REMIT_CLAIM; source Invoice; cost component resale; views `RESELLER_MARGIN`(resale invoice) = resale − vendor amount and `REMITTANCE_MARGIN`(deal) = earnings − partner cost ± fx timing − fees. D24's forex line is the true-up of T50 and T51; D23 stands: own-desk

balances are ordinary partner-coded client accounts with the own-desk configuration as holder. *Amendment:* `INVOICE.vendor_invoice_id` pairs a resale invoice with the vendor invoice it marks up; `DEAL_GROUP.invoice_amount` (the part of the resale invoice this group collects, advance then balance) and `invoice_rate` (payout currency per invoice currency at pricing, the stored base of every reserved amount); `INVOICE_BALANCE` is a receivable balance for a resale invoice (amount – Σ invoice_amount collected) and a payable balance over released lines for a vendor invoice; invariant 23 restated for the two-invoice case; invariant 44 adds that a cash-rounding surplus is revenue, not a forex result.

5. **Earnings retention and offset.** `CONVERSION.residual_in` (already in the model) is the margin kept in the collection currency, with EARN_GROSS recognised in that currency; a table `EARNINGS_OFFSET` (project, from partner config and currency and amount, to partner config and currency and amount, the market rate or partner rate version used, the EARNINGS_RECEIVABLE rows recovered, requested_by Finance, approved_by Management, state) as the source of an EarningsOffset transaction; `EARNINGS_RECEIVABLE.recovered_by_offset_id` beside recovered_by_deal_id; ROLE_RIGHT action earnings_offset. Invariant 16 adds the offset as a movement of ownership in both directions at one stated rate.
6. **NPL's own currency exchanges.** Purpose FX_RESULT (holder None; amendment, replacing FX_CLEARING): a profit-and-loss account per currency — currency NPL sold is a debit, currency it bought a credit — for an earnings offset, a pool squared from NPL's USDT, a rebate, share or loss share paid in another currency, or NPL-GR's EUR claim paid to its wallet in USDT. Its reporting-currency value is NPL's realised currency result; it needs no closing. (FX_CLEARING, an asset, could never be closed under per-currency balancing, since only an income or expense line in each currency can absorb it.)
7. **Views over postings.** `LEDGER_POSTING.receiver_group_id` on every PAYABLE line; `PROJECT_BALANCE` = the DUE account balance per partner; `GROUP_ENTITLEMENT` = Σ PAYABLE postings by group; invariant 7 restated over postings (amendment: the v5.3 definitions summed deals and no longer agreed with the accounts).
8. **Payout mechanics.** INTRANSIT retired for client money: the promise is discharged on release (SETTLEMENT released_at), re-raised by SETTLEMENT_RETURN or a short CONFIRMATION; invariant 7 already subtracts sent lines from the entitlement, so the ledger now matches it; `SETTLEMENT_LINE.outcome` gains the value short, so "every released line ends confirmed, short or returned" (invariant 40) is a check on the column (v5.9). `CURRENCY_PAIR.amount_rounding` gains the value exact (the default for bank transfers; the unit options apply to cash hand-overs). A pool may be negative between a top-up NPL funded and the next conversion, which then includes NPL's own USDT (a conversion with a company part, posted through FX_RESULT).
9. **Fee arithmetic.** Invariant 42: fee parts are percentages of the principal at the basis rate, never compounded; *amendment:* in amount_mode target_out the fee is a markup on the target — amount_in = target × (1 + sender_pays_pct) ÷ rate, client net = target × (1 – receiver_pays_pct) plus any cash-rounding surplus — NPL's own quoting convention, which the ERD had left unstated.
10. **Holder types.** `LEDGER_ACCOUNT.holder_type` keeps Sender for LOSS_RECEIVABLE and RESALE_RECEIVABLE and Receiver for VENDOR_PAYABLE; the own desk is a PARTNER_CONFIG, not a new type.

Draft v5.9 — for review with NPL. Figures are worked examples, not NPL data, except the invoice amounts and instalment rates of Pattern M, which are NPL's.